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A systematic review of ketamine and esketamine-induced long-term potentiation and synaptic scaling: Do the molecular and synaptic plasticity effects inform dosing intervals?

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ABSTRACT

Ketamine and esketamine exhibit rapid antidepressant effects in persons with treatment-resistant depression (TRD) and bipolar depression (TRBD). However, the synaptic mechanisms governing dose, frequency, and durability of response remain unclear. This review aims to evaluate the dosing parameters, including minimum dose and optimal dosing intervals, that reliably induce or maintain long-term potentiation (LTP) and/or synaptic scaling in humans and preclinical models. A systematic search (database inception to June 2025) was conducted on OVID and PubMed to identify preclinical and clinical studies reporting ketamine-induced clinical symptom changes alongside direct or indirect marker of LTP and/or synaptic scaling. Study selection, quality assessment, and data extraction were conducted by two independent reviewers. Sixty-one clinical and 17 preclinical studies met inclusion criteria. Most clinical studies enrolled persons with TRD, with fewer including TRBD or mixed TRD/TRBD samples. In TRD, a single 0.5 mg/kg intravenous ketamine infusion produced rapid but transient antidepressant effects, peaking at 24-hours and declining over 2–3 days. A similar temporal pattern was observed in TRBD. Early neurophysiological changes emerged within 3–8-hours, consolidated by 24-hours, and were sparsely detected beyond 3-days post-ketamine treatment. Consistent neurophysiological findings were observed in preclinical models. Across clinical trials, twice- versus thrice-weekly dosing yielded comparable four-week outcomes, and weekly maintenance significantly reduced relapse risk. Ketamine may open a plasticity window lasting approximately 2–3 days. Dosing intervals aligned with this window during an acute course of treatment, informed by rapid neurophysiological markers, may optimize antidepressant durability and response while minimizing drug exposure.

1. Introduction

Major depressive disorder (MDD) and bipolar disorder (BD) are persistent and debilitating heterogeneous mood disorders, projected to become the leading cause of disability by 2030, worldwide (Bains and Abdijadid, 2025; GBD 2021 Causes of Death Collaborators, 2024; GBD 2021 Nervous System Disorders Collaborators, 2024; Maj et al., 2020). MDD and BD share core features including sustained/depressed mood, anhedonia, impaired cognition, and altered occupational and social functioning, yet differ significantly in disease course, comorbidities, and treatment responsiveness (Maj et al., 2020; Wu et al., 2025; McIntyre et al., 2013; Huang et al., 2023; Wilkowska et al., 2024; Yang et al.,

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2023; Krishnan, 2003; Zheng et al., 2019; Inoue et al., 2020; Vázquez et al., 2011; Ghaemi et al., 2004; Perugi et al., 2019; Lee et al., 2018; McIntyre et al., 2020a).

Treatment-resistant depression (TRD), affecting ~6–55 % of individuals in real-world clinical practice, is defined as failure to respond to two or more antidepressant regimens despite adequate dosage, duration, and adherence to treatment (McIntyre et al., 2023; Thomas et al., 2013; Liu et al., 2021; Nemeroff, 2007; Clinical investigation of medicinal products in the treatment of depression - Scientific guideline, 2013; Research C for DE and. Major Depressive Disorder, 2020; Gaynes et al., 2008). Similarly, roughly a third of individuals with BD meet criteria for treatment-resistant bipolar depression (TRBD) (Sienaert et al., 2013; Gitlin, 2006; Hidalgo-Mazzei et al., 2019; McIntyre and Calabrese, 2019). Low remission rates associated with current first-line antidepressant treatments suggest complex neurobiological underpinnings beyond neurotransmitter imbalances (i.e., serotonin and dopamine) (Hidalgo-Mazzei et al., 2019; Caraci et al., 2018; Mancuso et al., 2023; Yalin and Young, 2020; Jiang et al., 2022; Ashok et al., 2017; Dembek et al., 2023; Stimmel, 2004; McIntyre et al., 2020b; Sussman et al., 2019; Pérez-Sola et al., 2021).

Racemic ketamine and esketamine (i.e., *N*-methyl-D-aspartate (NMDA) receptor antagonists), are established as efficacious pharmacotherapeutics in TRD, with preliminary evidence of rapid and robust antidepressant efficacy in TRBD (Moghaddam et al., 1997; d'Andrea et al., 2023; Wong et al., 2024; Wang et al., 2024; Berman et al., 2000; Diazgranados et al., 2010a; Canuso et al., 2018; Daly et al., 2018a; Murrough et al., 2013a; Zarate Jr et al., 2006; McIntyre et al., 2021; Jawad et al., 2023; McIntyre et al., 2025). Several systematic reviews and meta-analyses indicate dose-dependent efficacy of intravenous (IV) racemic ketamine (0.2–1.0 mg/kg) and intranasal (IN) esketamine (56-84 mg twice-weekly) for TRD (Fountoulakis et al., 2025; Wilkowska and Cubała, 2023; Fancy et al., 2023a; Bahji et al., 2021a; Nikolin et al., 2023; Seshadri et al., 2024; Calder et al., 2024). In addition, effectiveness in real-world populations with these treatments is also being established (d'Andrea et al., 2024; McIntyre et al., 2020c). Furthermore, adjunctive IV ketamine (0.5–0.75 mg/kg) has also demonstrated antidepressant efficacy as well as improved quality of life and functioning for persons living with TRD and TRBD (Fancy et al., 2023a; Cheng et al., 2025; Vieta et al., 2025). Notwithstanding replicated efficacy and effectiveness data for ketamine, optimal dosing frequency and treatment duration remains uncertain, and current clinical practices are centered on clinical endpoints. The foregoing gap in literature instantiates a need to determine optimal dosing intervals as well as dosing range based on neurobiological mechanistic insights (i.e., how long drug-induced antidepressant biology persists after each dose) (Smith-Apeldoorn et al., 2022; Phillips et al., 2020a; Nijs et al., 2020).

Pre-clinical and translational research suggest ketamine's rapid antidepressant effects may involve synaptic plasticity processes, including Hebbian long-term potentiation (LTP) and/or homeostatic scaling which differ in their location, time course, and mechanisms of effect (Krystal et al., 2024; Kavalali and Monteggia, 2020). At the molecular level, LTP is a persistent strengthening of a neuronal synaptic connection post-repeated activation, resulting in increased synaptic efficacy which takes minutes to hours and is crucial for learning and memory (Nicoll, 2017; Abraham et al., 2019). On a large-scale network level, synaptic scaling is a homeostatic process wherein neuronal excitability is adjusted to maintain stable firing rates and network stability (Krystal et al., 2024; Turrigiano, 2012; Wu et al., 2021; Turrigiano, 2008a). Synaptic up-scaling observed post-ketamine/esketamine is potentially dependent on brain-derived neurotrophic factor (BDNF) signaling through its receptor tropomyosin receptor kinase B (TrkB). Specifically, NMDA receptor blockade alleviates eukaryotic elongation factor-2 (eEF2) inhibition, triggering dendritic BDNF release which activates mammalian target of rapamycin (mTOR) and extracellular signal-regulated kinase (ERK) pathways (Li et al., 2010; Autry et al., 2011). mTOR-ERK cascades are critical for LTP and synaptic scaling since they drive local translation and cytoskeletal remodeling (e.g., GluA1, PSD-95, synapsin) which consolidate newly potentiated and/or up-regulated synapses, and enlarge spines (Hoeffer and Klann, 2010; Henry et al., 2018). It is posited that if synaptic scaling—or other synaptic plasticity changes—emerge, and persist for weeks, frequent dosing may be unnecessary; however, rapid decay could necessitate adjusted dosing intervals (Phillips et al., 2020a; Phillips et al., 2019a; McInnes et al., 2022).

Existing reviews have not yet examined whether the dosage or inter-dose interval of ketamine/esketamine can be set by mechanistic evidence of Hebbian LTP and/or homeostatic synaptic scaling. To address this gap, this systematic review collates clinical and preclinical studies that not only evaluates antidepressant outcomes but also couples them with direct or indirect surrogate synaptic plasticity markers (e.g., electroencephalography (EEG) spectral signature shifts, structural and functional magnetic resonance imaging (MRI), indices of synaptic scaling, and electrophysiological LTP paradigms) (Le et al., 2024). The primary objective is to precisely estimate the relationship between dosing and measures of LTP and/or synaptic scaling, with an overarching aim to inform and support point-of-care decisions in adults receiving ketamine-/esketamine-based treatment for TRD or TRBD.

2. Methods

2.1. Search strategy

The systematic review followed the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). Relevant articles were retrieved from OVID (MEDLINE, Embase, AMED, PsychINFO, JBI EBP Databases) and PubMed from database inception through June 2025, using an adapted search string incorporating terms related to ketamine, esketamine, and neurophysiological mechanisms (e.g., LTP and synaptic scaling; detailed search string is presented in Table 1) (Seshadri et al., 2024). No search filters were applied. A manual search on Google Scholar and references of pertinent studies were also screened to ensure all relevant articles were identified.

2.2. Study selection and inclusion criteria

Covidence platform was used for duplicate removal and article screening, which was conducted by two independent reviewers (G.H.L. and S.W.) (Covidence, n.d.). Articles deemed potentially relevant by at least one reviewer during title and abstract screening underwent full-text review according to the predefined inclusion and exclusion criteria (Table 2). Discrepancies were resolved via discussion, with final inclusion determined by consensus between the reviewers. Studies investigating only ketamine/esketamine metabolites (e.g., hydroxynorketamine) were excluded as exogenous metabolites may not accurately reflect in vivo ketamine/esketamine-induced synaptic plasticity. Studies were also excluded if subanesthetic ketamine/esketamine doses were administered during anesthetic treatment regimens. In addition, studies investigating subcutaneous administration of ketamine/esketamine were excluded a priori to maintain methodological homogeneity as the review focused on the routes most widely studied and clinically implemented in TRD and TRBD. The full inclusion and exclusion criteria applied in this review are detailed in Table 2, which outlines the population, intervention, comparison, outcome, and study-design parameters used to guide screening. These criteria were applied consistently across all stages of study selection.

The diagnostic scope of this review was intentionally limited to MDD, bipolar depression, TRD, and TRBD, as these are the clinical populations in which ketamine-related neuroplasticity markers have been investigated. Other depressive disorders (e.g., seasonal affective disorder, premenstrual dysphoric disorder, postpartum depression) were outside the prespecified inclusion criteria and therefore not

Table 1
Search String.

("bipolar affective disorder" OR "bipolar affective psychosis" OR "bipolar depression" OR "bipolar disorder" OR "bipolar illness" OR "bipolar mood disorder" OR "BD" OR "depression" OR "depressive disease" OR "depressive illness" OR "depressive neuros*" OR "depressive personalit*" OR "depressive psychos*" OR "depressive state*" OR "depressive symptom" OR "depressive syndrome" OR "depressive disorder" OR "manic depression" OR "manic disorder" OR "manic depressive" OR "seasonal affective disorder" OR "seasonal mood disorder" OR "major depression" OR "major depressive disorder" OR "major depress*" OR "MDD" OR "long-term depression" OR "LTD")
AND
("antidepressant-resistant" OR "anti-depressant-resistant" OR "antidepressant resistant" OR "refractory" OR "therapy-resistant" OR "therapy resistant" OR "treatment-resistant depression" OR "treatment resistant depression" OR "TRD") AND ("arketamine" OR "esketamine" OR "ketamine" OR "racemic ketamine" OR "r-ketamine" OR "s-ketamine" OR "spravato")
AND
("neuroplasticity" OR "neuronal plasticity" OR "long-term potentiation" OR "LTP" OR "synaptic scaling" OR "homeostatic plasticity")
AND
("frequency" OR "dosing" OR "dose" OR "administration schedule" OR "dose interval" OR "administraction frequency" OR "treatment schedule" OR "dose response" OR "interval" OR "efficacy" OR "safety").

Table 2
Eligibility criteria for screening and determining relevance of articles.

Inclusion Criteria	1. Primary (clinical studies) or post-hoc studies 2. Sample consisting of adult human participants (aged 18–65) or healthy, depression/chronic-stress, and/or acute-stress animal models (pre-clinical) 3. Human participants with a primary diagnosis of major depressive disorders (MDD), bipolar depression (BD), treatment-resistant depression (TRD), or treatment-resistant bipolar depression (TRBD) 4. Investigates the efficacy and safety of ketamine (sub-anesthetic doses) and/or esketamine, and Hebbian LTP, homeostatic scaling, or their durability (optional for human studies, required for animal/pre-clinical studies) 5. Studies must report dosage and/or dosing frequency of ketamine and/or esketamine 6. Studies must measure outcomes of interest pre-post treatment 7. Full-text article available online 8. English language (or translated to English)
Exclusion Criteria	1. Secondary research articles (e.g., literature reviews, systematic reviews, meta-analyses, posters, abstracts, guidelines, protocols, etc.) or case-studies/series, and preliminary/interim analyses of ongoing clinical trials 2. Only measures baseline or endpoint outcomes (e.g., predictability studies) 3. Subanesthetic ketamine/esketamine administration during anesthetic treatment regimens 4. No investigation of the efficacy, safety, synaptic plasticity (e.g., Hebbian LTP, homeostatic scaling, or their durability) effects of ketamine and/or esketamine 5. Compares ketamine and/or esketamine with another pharmacotherapeutic or treatment strategy or augmentation without reporting ketamine/esketamine only findings related to efficacy, safety, and/or synaptic plasticity effects 6. Investigates only ketamine/esketamine metabolites 7. Mixed participant sample (e.g., 30 % MDD, 30 % BD, 30 % GAD, etc.) without reporting separate results specific to each of the depressive disorders listed in the inclusion criterion 3 8. Participant sample with a diagnosis that is not the main depressive disorders listed in the inclusion criterion 3 9. Animal models of other disorders that are not listed in the inclusion criteria 10. Full-text is not available online or article retracted

considered.

2.3. Data extraction

Two independent reviewers (G.H.L. and S.W.) conducted data extraction, using a piloted data extraction template, and discrepancies were solved through discussion. The information to be extracted from the studies was established a priori. For human studies, the information to be extracted included 1) author(s), 2) study design, 3) diagnosis, 4) sample size, 5) dosage, 6) dosing frequency, 7) efficacy and safety outcome(s), 8) synaptic plasticity outcome(s). Information regarding synaptic plasticity outcomes were only recorded when applicable.

The information extracted from preclinical/animal model studies were the same as for human studies; however, the "efficacy and safety outcome(s)" were relabeled as "behavioral outcome(s)" and synaptic plasticity outcome(s) were required. Synaptic plasticity outcomes include direct and indirect evidence of the occurrence of LTP and/or synaptic scaling. Direct markers of synaptic plasticity includes electrical responses of neurons to stimuli pre-post LTP induction, changes in synaptic structure (e.g., postsynaptic spine structure or density), and pre- and post-synaptic alterations (Glasgow et al., 2019; Blundon and Zakharenko, 2008; Schulz et al., 1999; Fedulov et al., 2007; Turrigiano, 2008b; Canals et al., 2009). Indirect markers include changes in synaptic strength and receptor trafficking, as well as correlations between various forms of synaptic plasticity and neuronal activity (Turrigiano, 2008b; Canals et al., 2009; Desai et al., 1999; Perrett et al., 2001; Castillo, 2012; Guntupalli et al., 2016).

2.4. Quality assessments

The Quality Assessment of Controlled Intervention Studies, adapted from the National Institute of Health (NIH), was applied to assess the risk of bias (RoB) of the included studies with human participants (Study Quality Assessment Tools, n.d.; Ma et al., 2020). Quality assessment for pre-clinical/animal studies was conducted using SYRCLE's RoB tool which is an adapted version of the Cochrane RoB tool for randomized clinical trials (Hooijmans et al., 2014). Quality assessments of the included studies were conducted by two independent reviewers (G.H.L. and A.L.) and conflicts were resolved through discussion (Tables S4-S7).

3. Results

3.1. Search results

The systematic search resulted in the identification of 335 articles. A manual search on Google Scholar and references of published systematic reviews/meta-analyses resulted in the identification of 48 additional articles. After the removal of duplicates, 301 articles underwent title and abstract screening. During title and abstract screening, secondary research articles identified immediately were also removed. Subsequently, a total of 121 relevant articles underwent full-text screening in accordance with the eligibility criteria, and 31 articles were excluded (Fig. 1). In total, 90 articles (73 clinical studies and 17 preclinical studies) were retrieved and underwent data extraction. Study characteristics and results are described in supplementary materials Tables S1-S3.

3.2. Methodological quality

Across all 73 clinical studies, there were 47 controlled intervention studies, 21 pre-post studies with no control group, and 5 observational cohort or cross-sectional studies. Controlled intervention studies generally showed a low risk of bias. In contrast, pre-post studies with no control group and observational and cross-sectional studies tended to

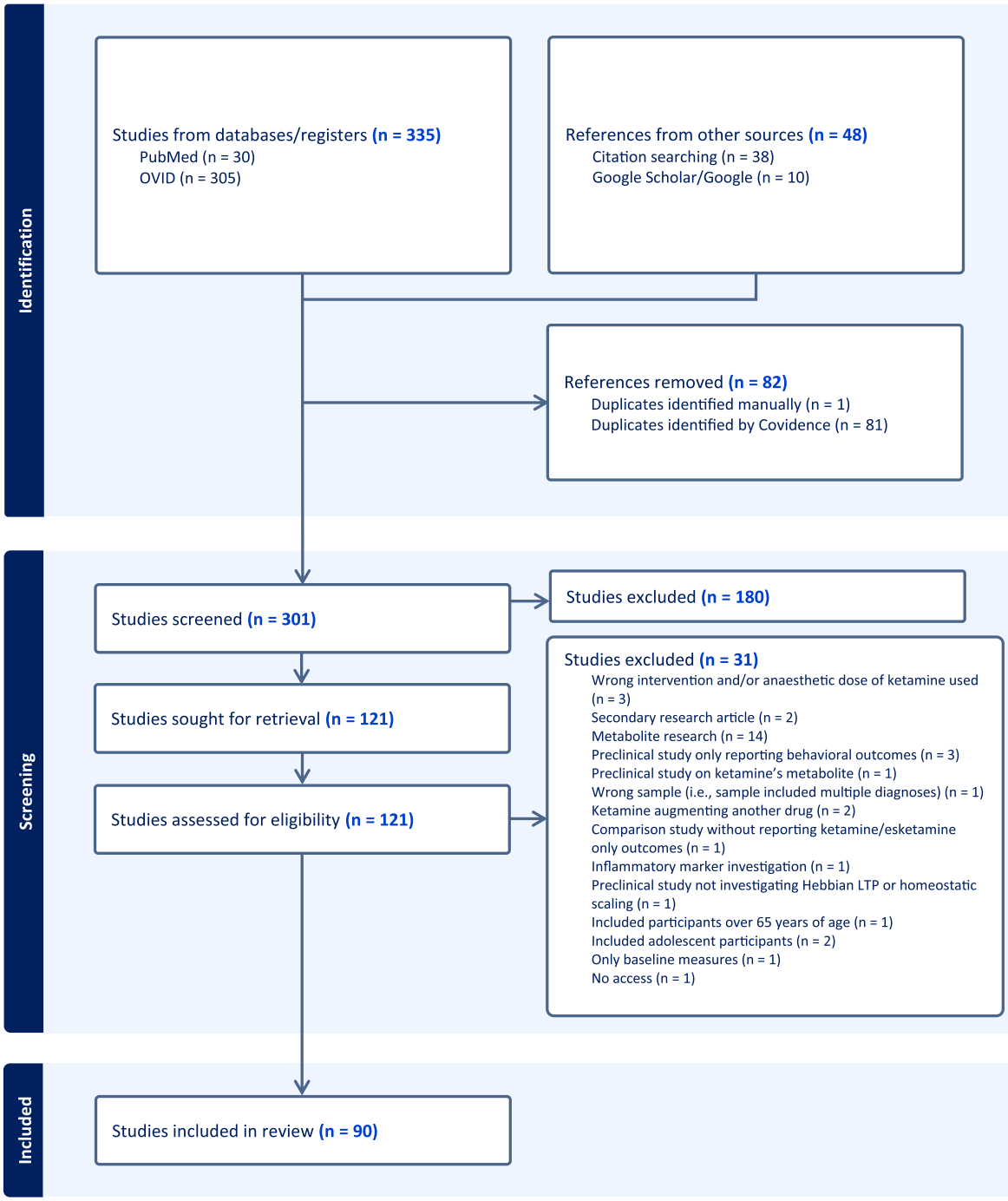

Fig. 1. PRISMA flow diagram of literature search (Covidence, n.d.).

exhibit a moderate or low risk of bias. Across all 17 preclinical studies, a moderate risk of bias was mostly found; however, a lack of reporting of several study design details largely contributed to these evaluations. Consequently, all preclinical studies were still included in the main analysis. See supplementary materials for further details regarding the quality assessment outcomes (Tables S4-S7).

3.3. Clinical efficacy & dosing-frequency profile

We retrieved a total of 60 primary research studies that investigated the efficacy and safety of IV racemic ketamine ($n = 25$ single-dose with $n = 1$ investigating single-dose IV ketamine versus IV esketamine; $n = 10$ repeat-dose; $n = 7$ both single and repeat-dose), IN racemic ketamine (n = 1), oral racemic ketamine ($n = 4$), IN esketamine ($n = 12$), or IV esketamine (n = 1 only IV esketamine) for TRD and/or TRBD (Table S1). Converging evidence indicates ketamine and esketamine consistently demonstrated rapid antidepressant and antisuicide effects in TRD and TRBD (Table S1).

Across clinical studies, several trials reported transient adverse events including, but not limited to, psychotomimetic and/or dissociative symptoms (e.g., perceptual disturbances, derealization, and thought disorganization), reflecting ketamine and esketamine's well-established acute effects (Table S1). Adverse events were transient and resolved within 30–120 min post-treatment and did not progress to persistent psychosis, even in multi-session or multi-week treatment protocols. No included clinical study reported treatment emergent or sustained

psychotic symptoms.

3.3.1. Single-dose IV racemic ketamine

Intravenous racemic ketamine at a 0.5 mg/kg subanesthetic dose was consistently identified as a robust efficacy threshold (Li et al., 2016; Milak et al., 2020; Fava et al., 2020; Chen et al., 2018; Su et al., 2017), with significant depressive symptom relief observed as early as 110-minutes (110 m) after a single infusion and peaking at 24-hours (24 h) (Cohen's d = 1.46, 95 % confidence interval (CI) [0.91, 2.01]) (Murrough et al., 2013a; Zarate Jr et al., 2006; Li et al., 2016; Milak et al., 2020; Fava et al., 2020; Chen et al., 2018; Su et al., 2017; Rybakowski et al., 2017; Salvadore et al., 2012; Thakurta et al., 2012; Salvadore et al., 2010; DiazGranados et al., 2010; aan het Rot et al., 2010; Salvadore et al., 2009; Price et al., 2009; Su et al., 2023a; Chen et al., 2021; Phillips et al., 2020b; Phillips et al., 2019b; Grunebaum et al., 2017; Zarate et al., 2012; Ibrahim et al., 2012; Valentine et al., 2011; Diazgranados et al., 2010c; Su et al., 2023b; Correia-Melo et al., 2020; Tiger et al., 2020). While most randomized controlled trials (RCTs) reported robust antisuicide effects, two studies (Grunebaum et al., 2017; Nierenberg et al., 2023) observed nonsignificant improvements compared to placebo 24 h and 48 h post-infusion (Table S1) (Price et al., 2009; Su et al., 2023a; Zarate et al., 2012; Diazgranados et al., 2010c; Grunebaum et al., 2018). Significant antidepressant and antisuicide effects were maintained for up to 7-days post-infusion for some participants; however, there was an average trend towards small increases in depressive symptoms severity for every additional day post-infusion (b = 0.0004, 95 % CI [0.00009, 0.00062]), returning towards baseline levels after day-3 (mean HAM-D scores; baseline = 22.55 ± 5.2, Day-7 = 20.94 ± 6.93, $p > 0.05$) (Murrough et al., 2013a; Zarate Jr et al., 2006; Rybakowski et al., 2017; Thakurta et al., 2012; Phillips et al., 2020b; Zarate et al., 2012; Valentine et al., 2011). Only one study observed sustained improvements in MADRS scores >7-days post-ketamine compared to placebo (i.e., during 1-month and 2-month post-ketamine follow-ups, $p < 0.001$) (Zolghadriha et al., 2024). Dose-response evaluations confirmed minimal to no additional improvements beyond doses of 0.75 mg/kg ($p > 0.05$), suggesting a ceiling effect at higher dosages (i.e., 1.0 mg/kg) (Milak et al., 2020; Fava et al., 2020; Su et al., 2017).

3.3.2. Repeat IV racemic ketamine

Across the r-dose studies, cumulative antidepressant effects were observed with each additional infusion and often plateaued between the fourth and sixth treatments (Table S1) (Berman et al., 2000; McIntyre et al., 2020c; aan het Rot et al., 2010; Chen et al., 2021; Phillips et al., 2020b; Phillips et al., 2019b; Singh et al., 2016a; Shiroma et al., 2020; Murrough et al., 2013b; Rasmussen et al., 2013; Zheng et al., 2018; Wilkowska et al., 2021; Fancy et al., 2023b; Chen-Li et al., 2025; Talaei et al., 2023; Ohtani et al., 2024; Pattanaseri et al., 2024). Furthermore, in comparison to a single infusion (response rate = 35 %), greater response (72 %) and remission (48 %) were observed post-4 ketamine infusions (Tiger et al., 2020). Singh et al. (2016a) reported significantly greater MADRS score reductions after twice-weekly (mean-change = −18.4, $p < 0.001$) and thrice-weekly (mean-change = −17.7, $p < 0.001$) ketamine infusions (i.e., 0.5 mg/kg) compared to placebo, with onset of clinical response after 2–3 infusions and maximal effects observed by days 15–19 (Table S1) (Chen et al., 2021; Phillips et al., 2019b; Singh et al., 2016a). Symtpom improvements were similar between twice and thrice-weekly ketamine dosing frequency groups at week-2 (i.e., after four and six ketamine infusions, respectively) and week-4 (i.e., after eight and 12 ketamine infusions, respectively) (Table S1) (Singh et al., 2016a). Furthermore, onset of clinical response was observed within the first week of ketamine infusions in both dosing frequency groups (Singh et al., 2016a). Maintenance infusions (i.e., weekly infusions for four weeks) stabilized symptom improvements (F = 0.88, $p = 0.49$); however, median time-to-relapse occurred ~18 days post-acute treatment phase (Table S1) (aan het Rot et al., 2010; Phillips et al., 2019b; Murrough et al., 2013b; Rasmussen et al., 2013; Zheng et al., 2018). Moreover, dose optimization strategies (i.e., increasing from 0.5 mg/kg to 0.75 mg/kg after two infusions with the index dose) significantly enhanced antidepressant (treatment-by-time interaction: F = 26.6, $p < 0.0001$, $\eta p2 = 0.16$) and antisuicide (F = 25.4, $p < 0.0001$, Cohen's f = 0.46) efficacy, with the greatest improvements in suicidal ideation severity post-third infusion (mean Columbia-Suicide Severity Rating Scale (C-SSRS) change = −1.1, 95 % CI [−1.4, −0.8], $p < 0.001$) (McIntyre et al., 2020c; Fancy et al., 2023b; Chen-Li et al., 2025).

3.3.3. Ketamine and esketamine with different routes of administration

Intranasal ketamine ($n = 1$) and esketamine ($n = 11$) produced significant rapid antidepressant effects within 40 m ($p < 0.001$) to 4 h post-dose (least-square (LS)-mean difference = −5.4, Cohen's d = 0.61, $p < 0.05$) (Table S1) (Canuso et al., 2018; Daly et al., 2018a; Lapidus et al., 2014; Fu et al., 2020). Although dosing strategy was not reported (Table S1), significant reductions in depressive symptom severity were observed up to 3-months post-IN esketamine treatment (Martinotti et al., 2022; Martinotti et al., 2023). Repeat fixed-dose IN esketamine (28-84 mg) consistently demonstrated significant dose-response effects on depressive symptom improvements (Canuso et al., 2018; Daly et al., 2018a; Takahashi et al., 2021; Fedgchin et al., 2019). Mixed findings were observed for flexibly-dosed IN esketamine wherein one study reported significant symptom improvements lasting up to day-28 (LS-mean difference = −4.0, 95 % CI [7.31, −0.64]) (Popova et al., 2019), while others did not report on statistical significance or observed nonsignificant differences between esketamine and placebo treatments (Chen et al., 2023; Ochs-Ross et al., 2020; Wajs et al., 2020). Maintenance IN esketamine treatment significantly delayed relapse among stable responders (hazard ratio (HR) = 0.30, 95 % CI [0.16, 0.55], $p < 0.001$) and remitters (HR = 0.49, 95 % CI [0.29, 0.84], $p = 0.003$) (Wajs et al., 2020; Daly et al., 2019).

Similarly, oral ketamine (0.5-1 mg/kg) demonstrated rapid antidepressant effects as early as 40 m post-dose (1 mg/kg oral ketamine, $p = 0.045$) though available evidence remains limited (Arabzadeh et al., 2018; Domany et al., 2019). Furthermore, greater depressive symptom improvements were observed for higher doses (i.e., 120-180 mg) compared to lower doses (i.e., 30-60 mg); however, there were no significant differences when compared to >180 mg oral ketamine doses (Table S1) (Glue et al., 2024; Colla et al., 2024). In addition, we also obtained two studies on IV esketamine (0.2 mg/kg: LS-mean changes in MADRS score = −16.8, one-sided $p = 0.001$; 0.4 mg/kg: LS-mean changes in MADRS score = −16.9, one-sided p = 0.001) wherein significantly greater symptom improvements were observed at 24 h compared to placebo in one study, while the other study observed peak remission rates at 72 h post-esketamine (35.5 %) (Correia-Melo et al., 2020; Singh et al., 2016b).

3.4. Effect of intravenous ketamine on long-term potentiation (LTP) and/or homeostatic/synaptic scaling in persons with treatment-resistant depression

This section presents the findings of 12 clinical studies investigating the effect of 0.5 mg/kg IV racemic ketamine on LTP and/or homeostatic scaling up to 72 h post-treatment in persons with TRD ($n = 6$ open-label, $n = 3$ double-blind RCTs, n = 3 double-blind crossover RCTs) (Table S2). No studies investigating this effect with esketamine, other subanesthetic ketamine doses, or other routes of administration were found.

3.4.1. Direct evidence of LTP and/or synaptic scaling after single-dose ketamine

Evidence indicates several direct physiological markers of Hebbian synaptic strengthening as early as 4-8 h post-one infusion of 0.5 mg/kg IV ketamine. Specifically, Cornwell et al. (2012) conducted an open-label study wherein magnetoencephalography (MEG) recordings were performed at baseline and post-ketamine (i.e., 6.5 h post-infusion) (Cornwell et al., 2012). During each recording, stimulus-evoked

somatosensory cortical (SSctx) excitability was measured via tactile stimulation of the left and right index fingers (Cornwell et al., 2012). In the responders group ($n = 9$, response measured 230 m post-infusion), significantly increased stimulus-evoked gamma band potentiation in the SSctx was observed 6.5 h post-infusion (Time effect: $F(1,8) = 12.19$, $p = 0.008$; Response status by time interaction effect: $F(1,18) = 8.38$, $p = 0.01$) (Cornwell et al., 2012). In contrast, there were no significant changes in nonresponders ($n = 11$) (Time effect: $F(1,10) = 1.51$, $p = 0.25$) (Cornwell et al., 2012).

Consistent with Cornwell et al. (2012), Ramakrishnan et al. (2019) demonstrated a reduction in dorsolateral prefrontal cortex (PFC) local mean field amplitude-area under the curve (LMFA-AUC) (i.e., proxy for homeostatic resetting preceding LTP) 4 h post-infusion ($p < 0.01$), suggesting an early homeostatic excitability-inhibition reset that is indicative of synaptic scaling (Ramakrishnan et al., 2019). At 24 h post-infusion, a slight increase in LMFA-AUC was observed but remained lower than baseline and scaled with reductions in MADRS and HAM-D scores ($p < 0.01$) (Ramakrishnan et al., 2019). Furthermore, a reduction in transcranial magnetic stimulation (TMS) evoked potentials' (TEPs) N100 component's - linked to GABAergic inhibition - peak-to-peak amplitude was also observed at 4 h and 24 h post-infusion ($p < 0.05$) (Ramakrishnan et al., 2019).

Complementing the foregoing findings, a separate double-blind crossover RCT conducted electroencephalography (EEG) recordings 3-4 h post-infusion with a sensory LTP paradigm to induce LTP-like enhancements of early visually evoked potential (VEP) components (i.e., P2) (Sumner et al., 2020). In the late post-tetanus (i.e., 35 m after photic tetanus/high-frequency stimulation) measurement block, a significantly larger P2 amplitude was observed at 165 ms post-ketamine compared to the placebo ($t = 3.94$, $p = 0.017$ family-wise error corrected (FWE-c); treatment-by-time interaction effect: $F(1,232) = 14.65$, $p = 0.049$ FWE-c), suggesting potential maintenance or amplification of Hebbian LTP by ketamine (Sumner et al., 2020). Furthermore, source analyses reveal the occurrence of time-dependent LTP in visual cortical areas (i.e., bilateral inferior temporal cortices and bilateral superior parietal cortices) after the visual tetanus (i.e., bilateral middle occipital gyrus (MOG) relays the visual information to the visual cortical areas) in both treatment groups, indicating that activity in these areas is attributable to LTP whereas ketamine may modulate the magnitude or durability/persistence of LTP (Sumner et al., 2020). No significant relationship was observed between change in MADRS scores at 24 h post-ketamine and the greater late post-tetanus P2 potentiation modulated by ketamine (i.e., LTP) (Sumner et al., 2020). Notwithstanding, Sumner et al. (2020) noted that this null finding should not discount the potential role of ketamine-related LTP in antidepressant efficacy due to limitations in the analyses that may have obscured a true association.

3.4.2. Indirect evidence of LTP and/or synaptic scaling after single-dose ketamine

Indirect neuroelectrical physiological proxies for Hebbian LTP and/or synaptic scaling were also observed wherein there may be a shift towards feed-forward dominance with ketamine. During the LTP paradigm, ketamine increased modulation of forward connections from the MOG to visual cortical areas, potentially supporting persistent synaptic potentiation and broader synaptic scaling processes (Sumner et al., 2020). There is also indirect evidence for early network reconfiguration within ~2 h post-ketamine treatment. Specifically, functional connectivity (FC) between the subgenual anterior cingulate cortex and amygdala was significantly reduced 2 h post-ketamine compared to placebo ($p = 0.0003$), while the ventral-striatum seed demonstrated broad, subthreshold gains to the thalamus, putamen and paralimbic cortex (pFWE = 0.009) (Burrows et al., 2023). Furthermore, ketamine also reduced executive control resting-state network coherence with the left hippocampus and right amygdala (pFWE = 0.042) (Burrows et al., 2023). These findings indicate rapid decoupling of a depression-linked limbic circuit and homeostatic up-scaling of striatal networks suggestive of early metaplastic re-weighting that precedes the consolidation effects observed at 24 h post-treatment (Burrows et al., 2023).

Furthermore, three separate studies suggest circuit-level consolidation by 24 h post-ketamine infusion in persons with TRD (Murrough et al., 2015; Abdallah et al., 2017; Kopelman et al., 2023). In an open-label, resting-state functional MRI study, TRD patients had significant reductions in PFC global brain connectivity with global signal regression (GBCr) compared to healthy controls at baseline (Abdallah et al., 2017). At 24 h post-infusion, normalization of right-lateral-PFC connectivity (baseline GBCr = -0.04 ± 0.10, post-ketamine GBCr = 0.39 ± 0.09, Cohen's d = 1.13) was observed in the TRD group, consistent with the foregoing evidence for Hebbian potentiation and synaptic scaling (Abdallah et al., 2017). Compared to nonresponders, ketamine responders had greater changes in right-lateral-PFC GBCr, as well as significantly greater GBCr in the bilateral caudate, bilateral-lateral-PFC and left middle temporal regions, suggestive of a distributed plasticity cascade consolidating antidepressant network rebalancing (Abdallah et al., 2017). Furthermore, strong network plasticity, as indexed by greater changes in GBCr, were associated with greater improvements in MADRS scores, and there was a significant treatment effect on depression symptom severity (change in MADRS = 14.2 ± 1.9, $t = 7.6$, df = 17, $p = 0.000001$) (Abdallah et al., 2017). In line with these findings, a separate open-label study (using functional MRI blood oxygenation level (BOLD) activity as a proxy for circuit engagement) reported significant re-engagement of the reward-hub (i.e., right-caudate) responses to positive emotion to levels observed in healthy controls (Murrough et al., 2015). Increased right-caudate connectivity was also significantly correlated with depressive symptom improvements 24 h post-ketamine infusion (Murrough et al., 2015).

In support of the foregoing trends, Kopelman et al. (2023) investigated mean diffusivity (MD) (i.e., measure of overall tissue structure and integrity as indexed by magnitude of water molecule diffusion) in three regions of interest including Broadman Area 10 (BA10), hippocampus, and amygdala pre-post-ketamine infusion in persons with TRD (Kopelman et al., 2023). Ketamine significantly improved MADRS total scores from baseline to 24 h post-infusion ($t(150) = -4.55$, $p < 0.001$). Compared to placebo, greater improvements in MADRS total score is associated with greater MD reductions in the left (Group effect: $\beta = 0.314$, $t(96) = 3.207$, $p = 0.002$; Group by MD-change effect: $\beta = 0.308$, $t(96) = 1.760$, $p = 0.082$) and right BA10 (Group effect: $\beta = 0.268$, $t(96) = 2.763$, $p = 0.007$; MD-change effect: $\beta = 0.196$, $t(96) = 2.019$, $p = 0.046$), and left amygdala (Group by MD-change effect: $\Delta R2 = 0.045$, $\beta = 0.414$, $t(96) = 2.238$, $p = 0.028$) 24 h post-ketamine infusion (Kopelman et al., 2023). In contrast, increased MD in the left and right hippocampus 24 h post-ketamine significantly predicted greater improvements in depressive symptom severity compared to placebo (Group by MD-change interaction effect; Left-Hippocampus: $\Delta R2 = 0.044$, $\beta = -0.337$, $t(96) = -2.178$, $p = 0.032$; Right-Hippocampus: $\Delta R2 = 0.033$, $\beta = -0.303$, $t(96) = -1.902$, $p = 0.060$) (Kopelman et al., 2023).

Several studies also reported molecular and serum changes indicative of Hebbian LTP- and/or scaling-related mechanisms. Compared to baseline, significant increases in plasma BDNF were observed by Duncan Jr et al. (2013) and Haile et al. (2014) at 230 m and 240 m post-ketamine, respectively (all p values <0.05) (Duncan Jr et al., 2013; Haile et al., 2014). In the latter study specifically, ketamine responders on day-7 had significantly greater plasma BDNF levels ($t = 2.450$, $p = 0.03$) and improvements in MADRS total score (group by time interaction: $F(5,86) = 3.93$, $p = 0.003$) than nonresponders, suggesting that a greater BDNF boost may seed downstream synaptic changes (e.g., LTP consolidation and homeostatic up-scaling) (Haile et al., 2014). Furthermore, a significant negative correlation between MADRS scores and plasma BDNF was observed in the ketamine ($F(1,12) = 10.646$, $p = 0.008$) but not midazolam ($F(1,5) = 0.235$, $p = 0.653$) group (Haile et al., 2014). In addition, plasma BDNF levels at 240 m is also a significant predictor of depressive symptom severity at 240 m-72 h ($p =$

0.001–0.038) but not 7-days ($p = 0.098$) post-infusion (Table S2), suggesting a potential plasticity window of ≤3 days wherein the initial burst of BDNF-triggered plasticity (e.g., AMPAR insertion, spine growth) begins to decay at 2–3 days (Haile et al., 2014).

In addition, two studies reported changes in metabolic correlates of antidepressant response after a single ketamine infusion in TRD and TRBD (Nugent et al., 2014; Evans et al., 2018). In the right ventral striatum at 200 m post-infusion, changes in regional metabolic rate of glucose (rMRGlu) are inversely correlated with percent improvement on MADRS total score ($r = -0.720$, $p < 0.001$) in persons with TRBD (Nugent et al., 2014). Whole-brain fluorodeoxyglucose (FDG) positron emission tomography (PET) analyses revealed significant reductions in left hippocampus glucose utilization ($t = 5.53$, adjusted $p < 0.001$), suggesting a rapid and transient dampening of limbic-hippocampal activity, which may reflect ketamine's facilitation of synaptic pruning within maladaptive cortico-striatal networks in preparation for subsequent Hebbian LTP and homeostatic scaling that supports clinical symptom improvement (Nugent et al., 2014). Separately, Evans et al. (2018) observed baseline-dependent bidirectional shifts in pregenual anterior cingulate (pgACC) glutamate levels after dosing; however, this was not significant (Evans et al., 2018).

We were only able to retrieve one study that attempted to investigate the persistence and durability of ketamine's effect on LTP and/or synaptic scaling beyond 24 h post-treatment. Using resting-state fMRI three days after ketamine infusion, they found post-infusion degree centrality (i.e., number of connections in a node) was significantly greater in the ketamine versus midazolam group in the left angular gyrus (39.4 ± 11.93 vs. 30.0 ± 8.46, respectively; $p = 0.005$) and right thalamus (51.4 ± 12.19 vs. 42.2 ± 13.49, respectively; $p = 0.023$) (Su et al., 2025). The same trends were observed for clustering coefficient values (i.e., likelihood of nodes forming triangles) in the ketamine versus midazolam groups in the left (0.16 ± 0.04 vs. 0.13 ± 0.03, respectively; $p = 0.004$) and right (0.16 ± 0.05 vs. 0.12 ± 0.04, respectively; $p = 0.006$) angular gyrus (Su et al., 2025). Although significantly greater reductions in depressive symptom and suicidality severity were observed 3-days post-ketamine infusion compared to midazolam (i.e., HDRS: $p = 0.011$, C-SSRS: $p = 0.013$), there were nonsignificant symptom-network correlations after correcting for multiple comparisons (all $p > 0.0042$) (Su et al., 2025).

3.4.3. Indirect evidence of LTP and/or synaptic scaling after repeat-infusions of ketamine

No studies provided direct evidence (e.g., event-related potentials, TMS metrics) of renewed Hebbian LTP and/or homeostatic scaling after repeat ketamine infusions (Table S2). Only one open-label study reported indirect evidence of the foregoing phenomena after four ketamine infusions (2–3 infusions/week) (Sahib et al., 2022). Compared to baseline, findings reveal a trend of increased functional connectivity within the somato-motor network towards healthy control levels 24 h after the first infusion and 24-72 h after the last infusion (Sahib et al., 2022). Furthermore, significant large-scale adjustments were observed 24-72 h post-fourth infusion, but not 24 h post-first infusion, including a significant decrease in functional connectivity between the cerebellum and salience network nodes, towards healthy control levels (Sahib et al., 2022). In addition, there was a significant remission status by time interaction effect on the functional connectivity between the cerebellum and striatum nodes ($F = 3.742$, $p = 0.027$), wherein the change from baseline to 24-72 h post-fourth infusion was negative in remitters and positive in non-remitters (Sahib et al., 2022). Significant improvements in HDRS scores were also observed across time ($F(1.89, 91.81) = 83.77$, $p < 0.0001$), with the greatest gains following serial infusions (Sahib et al., 2022). In line with the observed significant clinical improvement, Sahib et al. (2022) also reported a significant positive correlation between change in functional connectivity and %HDRS change from baseline to post-fourth infusion (Sahib et al., 2022).

3.5. Preclinical evidence of ketamine/esketamine-induced long-term potentiation (LTP) and/or synaptic scaling

Although preliminary clinical work provides evidence in support of ketamine-induced LTP and/or synaptic scaling in TRD, there is a lack of human studies that track these neuroplastic changes beyond 24-48 h post-treatment. To address this gap, we broadened our review to include preclinical studies ($n = 17$) that measure durability over longer intervals in wild-type and/or animal models of depression. Among the preclinical studies, two employed a treatment-resistant strain (i.e., Wistar-Kyoto rats), while all other models reflected standard depression-like paradigms or naïve rodents (Table S3) (Belujon and Grace, 2014; Aleksandrova et al., 2020). All preclinical studies meeting inclusion criteria employed antidepressant-relevant dosing paradigms and did not observe or report emergent or persistent psychosis-like behavioral abnormalities (Table S3).

Findings were similar across the preclinical studies that provided direct evidence of Hebbian LTP and/or synaptic scaling (Table S3). Within 1–24 h of treatment with a single sub-anesthetic dose of ketamine (i.e., 5-10 mg/kg intraperitoneal (IP) injection or 1–10 μM in brain slices) (and in one study; a subanesthetic dose of 2.5-10 mg/kg IP esketamine injection), findings suggest ketamine either induced, maintained, or restored LTP in the CA1 (i.e., critical area within the hippocampus crucial for learning and memory) or medial PFC, and/or increased dendritic spine density (Li et al., 2010; Nosyreva et al., 2013; Belujon and Grace, 2014; Graef et al., 2015; Widman et al., 2018; Yang et al., 2018; Aleksandrova et al., 2020; Ren et al., 2022; Lin et al., 2021; Zanos et al., 2023; Tang et al., 2020). Furthermore, in stress-induced mice (i.e., 100 μg/mL of corticosterone), field excitatory postsynaptic potential (fEPSP) strength was significantly increased post-ketamine and remained intact in the hippocampus 1 h post-drug washout ($p < 0.01$) (Piazza et al., 2024). Moreover, Graef et al. (2015) also reported significantly greater LTP magnitude 72 h post-treatment (vehicle-treated: 218.3 ± 16.3 % of baseline, $n = 5$ rats; ketamine-treated: 314.3 ± 20.1 % of baseline, $n = 6$ rats, $p < 0.05$), suggesting ketamine sustained the amplified potentiation up to at least 3-days post-treatment (Graef et al., 2015). Separately, another preclinical study observed that 3 μM ketamine did not significantly modify LTP induction or maintenance; however, ketamine doses of 10-100 μM significantly decreased and/or completely blocked LTP induction and maintenance (Table S3) (Ribeiro et al., 2014).

In accordance with the foregoing direct evidence, findings from preclinical studies investigating molecular and/or circuit surrogates of LTP and/or synaptic scaling suggest ketamine's effects may be rapid but transient (i.e., 2–3-day window). Within minutes of a subanesthetic dose of ketamine, there is rapid activation of the mechanistic target of rapamycin (mTOR) pathway in the medial PFC, and increased levels of phosphorylated and activated forms of extracellular signal-regulated kinases (i.e., ERK1 and ERK2) and protein kinase B (PkB/Akt); however, returning to basal levels 2 h post-treatment (Li et al., 2010). Consistent with the aforementioned trends, evidence also indicate rapid eukaryotic translation elongation factor 2 (eEF2) de-phosphorylation in parallel to a surge in BDNF levels in the hippocampus, and restoration of CaMKIIα-CREB–BDNF signaling in an LPS-induced depression model 30 m post-ketamine treatment; which was not observed 24 h later (Autry et al., 2011; Tang et al., 2020; Zanos et al., 2016). Furthermore, glutamatergic activity was significantly higher in ketamine (1.289 ± 0.334 F/F_0) compared to control cells (1.000 ± 0.278 F/F_0) ($p = 0.0022$) 1 h post-treatment (Zaytseva et al., 2023). By 24 h post-ketamine treatment, a significant increase in glutamate receptor subunits 1 and 2 (i.e., GluA1 and GluA2) in mouse hippocampal synaptoneurosomes were observed ($p < 0.05$) (Zanos et al., 2016). Circuit probes confirm a matching time-course wherein as soon as 0-10 m post-ketamine, bursting and overall spiking activity were significantly decreased in the lateral habenula (LHb) neurons of chronically stressed mice which lasted for at least 1 h ($p < 0.001$), suggesting an acute circuit reset via NMDA channel

trapping to rapidly normalize an over-potentiated hub (Chen et al., 2024).

4. Discussion

To our knowledge, this is the first systematic review to synthesize and extract literature on ketamine and esketamine with a focus on surrogate markers of Hebbian LTP and homeostatic scaling, integrating them with clinical outcomes, with a view to inform dosing intervals in TRD and TRBD. Consistent findings across the 50 clinical trials indicate that a single subanesthetic (i.e., 0.5 mg/kg) ketamine IV infusion produces rapid and significant antidepressant effects, peaking at ~24 h post-treatment and diminishing over the subsequent days (Murrough et al., 2013a; Zarate Jr et al., 2006; Thakurta et al., 2012; Phillips et al., 2019b; Zarate et al., 2012; Valentine et al., 2011). Clinical neurophysiological evidence follow a congruent trajectory wherein cortical disinhibition, homeostatic excitability-inhibition reset, and LTP amplification emerge within 4-8 h post-infusion (Cornwell et al., 2012; Ramakrishnan et al., 2019; Sumner et al., 2020). None of the clinical studies included investigated Hebbian LTP and/or synaptic scaling after a single ketamine infusion beyond the foregoing time-window; however, indirect evidence indicates normalization of fronto-limbic connectivity by 24 h (Murrough et al., 2015; Abdallah et al., 2017), and sparse network effects that were only observed up to 72 h post-treatment (Haile et al., 2014; Su et al., 2025).

A critical mechanistic link between LTP, homeostatic synaptic scaling and antidepressant efficacy is grounded in ketamine's ability to shift cortical networks from a maladaptive and low-plasticity, depression-associated state into a transiently enhanced plasticity state that supports large-scale, adaptive synaptic and circuit-level reorganization (Li et al., 2010; Autry et al., 2011; Hoeffer and Klann, 2010). Ketamine-induced disinhibition of pyramidal neurons results in an acute increase in excitatory drive that strengthens active synapses through Hebbian plasticity mechanisms, thereby enhancing the efficiency of information flow within cortico-limbic and cortico-striatal circuits implicated in mood regulation (Li et al., 2010; Autry et al., 2011; Hoeffer and Klann, 2010; Henry et al., 2018; Abdallah et al., 2017; Kopelman et al., 2023; Haile et al., 2014; Nugent et al., 2014). In parallel, based on our findings, we hypothesize that homeostatic synaptic scaling may operate as a compensatory, cell-wide regulatory process that modulates synaptic gain to maintain network stability (Cornwell et al., 2012; Ramakrishnan et al., 2019; Sumner et al., 2020). The foregoing suggests such scaling prevents excessive excitation, supporting the integration of Hebbian synaptic enhancements into a balanced and physiologically coherent circuit architecture (Ramakrishnan et al., 2019; Burrows et al., 2023).

Within this conceptual framework, Hebbian LTP confers the synapse- and circuit-level specificity necessary to reverse the maladaptive and pathological functional connectivity patterns observed in treatment-resistant depressive disorders. Separately, homeostatic scaling is posited to stabilize the LTP-enhanced synaptic networks over subsequent hours to days, thereby supporting the persistence of adaptive network dynamics and sustained improvements in affective processes and regulation of depressive symptomatology (Kavalali and Monteggia, 2020; Turrigiano, 2012; Wu et al., 2021; Turrigiano, 2008a; Murrough et al., 2015; Abdallah et al., 2017; Kopelman et al., 2023; Duncan Jr et al., 2013; Haile et al., 2014; Nugent et al., 2014).

To contextualize these mechanistic principles within the empirical literature, we next outline the temporal sequence of molecular, synaptic and electrophysiological findings observed across preclinical models. Within minutes to hours after a single subanesthetic dose of ketamine (5-10 mg/kg), rapid activation of mTOR-BDNF-eEF2 signaling is observed, as well as strengthening of fEPSP which remained intact in the hippocampus 1 h post-treatment in depression-like mice models (Li et al., 2010; Autry et al., 2011; Tang et al., 2020; Piazza et al., 2024; Zanos et al., 2016). This is followed by AMPAR insertion and dendritic spine growth, peaking at 24 h, and sustained enhancement of CA1 hippocampal LTP for ~48-72 h which begins to decay thereafter (Li et al., 2010; Haile et al., 2014; Nosyreva et al., 2013; Belujon and Grace, 2014; Graef et al., 2015; Widman et al., 2018; Yang et al., 2018; Aleksandrova et al., 2020; Ren et al., 2022; Lin et al., 2021; Zanos et al., 2023; Tang et al., 2020). These results demonstrate how the rapid induction and subsequent decay of plasticity-related signals mirror the conceptual framework described above.

Across the clinical and preclinical literature synthesized in this review, several conflicting findings were observed. For example, while most clinical studies reported robust neurophysiological changes within 24 h that decayed by 72 h (Murrough et al., 2015; Abdallah et al., 2017; Kopelman et al., 2023; Duncan Jr et al., 2013; Haile et al., 2014; Nugent et al., 2014; Su et al., 2025), a minority reported minimal or no changes beyond baseline, likely reflecting differences in assay sensitivity, sampling window, and/or patient-specific characteristics (McIntyre et al., 2020c; Covidence, n.d.; Murrough et al., 2015; Widman et al., 2018; Yang et al., 2018). Furthermore, sources of inconsistency include heterogeneity in depression models and animal strains, variability in behavioral paradigms, clinical heterogeneity in concomitant medications, illness severity, and chronicity (Tables S1-S3). These conflicting findings underscore the need for standardized mechanistic protocols and harmonized outcome measures to more precisely delineate dose-response relationships and temporal plasticity profiles.

Notwithstanding, the convergence of the aforementioned plasticity processes offers a mechanistic explanation for the temporal profile of ketamine's antidepressant effects, wherein rapid circuit reorganization is followed by a limited window of sustained stability, necessitating periodic re-engagement of plasticity mechanisms through repeated dosing. Furthermore, clinical and preclinical findings suggest that a single sub-anesthetic dose of ketamine or esketamine is able to not only produce rapid antidepressant and antisuicide effects, but also induce and sustain LTP and/or synaptic scaling. There is converging evidence of a 2–3-day plasticity window wherein synaptic potentiation and clinical symptom improvement overlap and begin to decay. While ketamine and esketamine can produce transient psychotomimetic effects, none of the included clinical or preclinical studies reported sustained psychosis-related outcomes under antidepressant-relevant dosing conditions. It is important to note that these findings are preliminary and hypothesis-generating thus future large-scale, adequately powered clinical trials replicating these findings are required to confirm the proposed dosing schedule.

4.1. Clinical Implications for dosing frequency to consolidate neuroplastic changes and enhance symptomatic improvement

In accordance with the current body of evidence, if synaptic potentiation subsides by the third day post-treatment, repeat 0.5 mg/kg IV ketamine infusions with a dosing interval of 2–3 days may preserve the neurobiological substrates associated with the observed antidepressant action in persons with TRD. Furthermore, the proposed dosing interval may also reduce excessive receptor engagement, and ultimately prevent potential receptor desensitization which could affect drug efficacy (Commons and Linnros, 2019). Separately, clinical trials directly comparing twice-weekly versus thrice-weekly dosing schedules report no administration frequency-related differences in overall symptomatic improvement (Singh et al., 2016a). Furthermore, across trials investigating repeat-dosing schedules observed a plateau in depressive symptom severity improvements and response rates after the 4–6 infusions (Shiroma et al., 2020; Wilkowska et al., 2021). Consequently, a twice-weekly dosing frequency for 2–3 weeks may be optimal to not only consolidate neuroplastic changes but also sustain antidepressant effects which is a pressing clinical need for treatment-resistant depressive disorders. Similar trends were observed for IN esketamine suggesting the proposed dosing schedule may be applicable. Specifically, twice-weekly for two weeks followed by weekly maintenance treatment with IN esketamine significantly reduced risk of relapse compared to placebo

(Daly et al., 2019; Daly et al., 2018b).

In support of the proposed dosing schedule, clinical evidence also indicates that the median number of infusions needed for treatment response is approximately three infusions, and the median time to relapse is 18 days post-six infusions (aan het Rot et al., 2010; Murrough et al., 2013b; Rasmussen et al., 2013; Zheng et al., 2018; Daly et al., 2019). Furthermore, real-world effectiveness retrospective studies also reported significantly greater improvements in depressive and suicidality symptom severity across time in TRD/TRBD persons receiving dose optimization (i.e., 0.5 mg/kg for the first two infusions and 0.75 mg/kg for the subsequent two infusions) compared to those that remained on the index (i.e., 0.5 mg/kg ketamine) dose (McIntyre et al., 2020c; Chen-Li et al., 2025). These findings also map onto the decay of neurophysiological mechanistic markers thus reinforcing the importance of both interval and dose optimization.

An additional observation that informs our proposed interval choice is the temporal coupling between early neurophysiological shifts and subsequent clinical symptom outcomes in persons with TRD or TRBD. For example, ketamine's early micro-structural effects may reduce the over-active and over-connected default mode network (i.e., interconnected brain regions most active at rest, and has been observed to be overactive and linked to tendency to ruminate in depression), ultimately reducing ruminative thoughts (Kopelman et al., 2023). Furthermore, post-ketamine somatosensory gamma potentiation at 6.5 h post-infusion may predict likelihood of response at 24 h (Cornwell et al., 2012). In parallel, a BDNF surge 4 h post-ketamine infusion has also been shown to significantly predict MADRS scores up to 3-days post-treatment (Haile et al., 2014). These findings suggest that these early changes, proxies of LTP and/or synaptic scaling induction and/or maintenance, post-ketamine treatment may not only inform dose and interval optimization but also may have potential predictive utility in identifying response to ketamine and/or esketamine as currently there is no fit-for-purpose biomarker prediction of treatment response (Rong et al., 2018).

4.2. Limitations and future directions

Our systematic review is not without methodological limitations that may affect inferences and interpretations of our findings. The clinical studies included in our systematic review primarily rely on surrogate markers rather than direct neurophysiological or molecular measures of synaptic strength, Hebbian LTP, and/or synaptic scaling. Moreover, there are challenges with respect to the translatability of preclinical findings to clinical human contexts. For example, brain slice findings in rodent models of depression are difficult to translate into steady-state cortical levels during a 40-minute human IV ketamine infusion. Consequently, these findings are hypothesis-generating and preliminary wherein future larger-scale and adequately powered clinical trials are required to compare response across different dosing frequencies, define optimal dosing schedules and determine whether neurophysiological changes coincide with clinical efficacy.

Furthermore, the sampling window of published trials is short wherein nearly all mechanistic recordings are terminated by 3-days post-treatment. Consequently, there is a lack of evidence with respect to the decay of LTP and/or synaptic scaling beyond the first week post-treatment. The foregoing limitation necessitates the investigation of long-term plasticity and antidepressant effects beyond the sampling window that has been reported. Such evaluations may contribute to our mechanistic understanding of ketamine's antidepressant effects, informing dosing schedules to enhance and sustain clinical symptom improvement as well as consolidate beneficial neurocircuitry alterations.

Secondly, while we aimed to evaluate both ketamine and esketamine, the current body of evidence remains disproportionately weighted towards racemic ketamine, limiting direct comparisons (Krystal et al., 2024). Notwithstanding, emerging data suggest several neurophysiological and clinical distinctions that may have implications for dosing optimization and schedules. Racemic ketamine contains equal proportions of the R- and S-enantiomers, with the R,S-ketamine increasingly implicated in longer-lasting synaptic and behavioral effects including more persistent activation of AMPA-BDNF-mTOR pathways and reduced psychotomimetic liability (Krystal et al., 2024; Hoeffer and Klann, 2010; Zhang et al., 2022; Bahji et al., 2021b; Singh et al., 2023; Kawczak et al., 2024). In contrast, esketamine has higher NMDAR affinity and more pronounced acute dissociative effects; however, may engage plasticity mechanisms through partially divergent downstream signaling cascades (Kawczak et al., 2024). Consequently, the optimal dosing intervals for ketamine and esketamine may differ. The distinctions between ketamine and esketamine underscore the importance of mechanistically informed, formulation-specific dosing strategies rather than assuming equivalence between agents that engage overlapping, but not identical, neuroplasticity pathways.

Moreover, there is heterogeneity in the methodology across trials. For example, there is heterogeneity regarding eligibility criteria (e.g., concomitant medications, primary diagnoses, age, sex, sample population, etc.) wherein each differing baseline variability between-subjects may be a potential mediator and/or modifier of glutamatergic plasticity (Kalivas and Volkow, 2011; Park and Bischof, 2013; Holmes et al., 2023; Rădulescu et al., 2021). Furthermore, individual-level clinical characteristics may also modulate response to ketamine and esketamine; however, the current mechanistic literature provides limited data to evaluate these effects. For example, age-related differences in glutamatergic signaling and synaptic plasticity have been described in both preclinical and clinical studies, raising the possibility that neurobiological aging may influence the magnitude or persistence of ketamine-induced LTP-like processes (Park and Bischof, 2013; Holmes et al., 2023; Pennybaker et al., 2021). Sex-related differences in baseline excitatory-inhibitory balance, hormonal modulation of BDNF-mTOR signaling and prevalence of treatment resistance may also affect response trajectories; however, there is a lack of mechanistic evidence stratified by sex (Chan and Ye, 2017; An et al., 2021; Scharfman and MacLusky, 2006; Autry et al., 2009; Young, 1998). In addition, number/trials of acute treatment steps with first-line monoamine-based antidepressants is another potentially important factor as there is evidence that more acute treatment steps is associated with lower acute remission rates and greater relapse rates during follow-up in MDD (McIntyre et al., 2023; Rush et al., 2006).

Separately, although unipolar and bipolar depression differ across multiple neurobiological dimensions (e.g., baseline glutamatergic tone, large-scale functional network organization, and mood-state-dependent plasticity dynamics), current literature and our findings indicate that ketamine and esketamine converge on a set of core antidepressant mechanisms that are conserved across depressive disorders (Yang et al., 2023; Ashok et al., 2017; McIntyre et al., 2021; Krystal et al., 2024; Li et al., 2010; Autry et al., 2011; Holmes et al., 2023). Given that mechanistic studies in TRBD remain comparatively limited, stratifying the synthesis of results strictly by diagnostic category would fragment the sparse evidence base without meaningfully altering the overarching mechanistic interpretation. Accordingly, we integrated findings across TRD and TRBD while noting diagnostic context where applicable and underscore the need for dedicated mechanistic investigations in bipolar depression to delineate any diagnosis-specific plasticity signatures.

Future clinical trials should venture into evaluating direct human assays of synaptic potentials beyond the current sampling window (i.e., $\geq$ 2-weeks post-treatment) via emerging tools such as high-density TMS-EEG paired associative stimulation and MEG paradigms, or PET tracers of ketamine-associated LTP and/or synaptic scaling (Esser et al., 2006; Stampanoni Bassi et al., 2019; Cai et al., 2019). In addition, instead of calendar-based redosing schedules, future trials should investigate dosing strategies that are informed by early predictive markers of response to tailor the timing of subsequent infusions to the presence of these signals in hopes of enhancing overall response rates and durability of antidepressant effects.

Future research vistas should also attempt to ascertain whether other antidepressant interventions that are glutamate modulators (i.e., dextromethorphan) also affect LTP and homeostatic scaling in a similar or different manner (McIntyre and Jain, 2024).

5. Conclusion

Convergent clinical, neurophysiological and preclinical evidence indicates that a single subanesthetic dose of ketamine is associated with induction of LTP and/or synaptic scaling as well as improvements in depressive symptom outcomes, peaking at ~24 h and subsiding over the subsequent 2–3 days. Dosing intervals of 2–3 days combined with weekly dose optimization may be sufficient to maintain and enhance the foregoing effects. Moreover, early molecular and neuroelectrophysiological changes may serve as real-time guides to improve personalized treatment strategies via adaptive, biomarker-triggered redosing schedules. Notwithstanding, we acknowledge that the evidence is preliminary and there is a critical need for longer-term investigations of LTP and synaptic scaling beyond the first week.

CRediT authorship contribution statement

Gia Han Le: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Sabrina Wong:** Writing – review & editing, Validation, Investigation, Data curation. **Danica E. Johnson:** Writing – review & editing, Validation. **Andy Lu:** Writing – review & editing, Data curation. **Diana Orsini:** Writing – review & editing. **Noah Chisamore:** Writing – review & editing. **Sara Di Luch:** Writing – review & editing. **Joshua D. Rosenblat:** Writing – review & editing, Supervision. **Roger S. McIntyre:** Writing – review & editing, Supervision, Project administration, Conceptualization.

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Appendix A. Supplementary data

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